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Roll No-

MID TERM EXAMINATION, M.TECH (AI)

MARCH-2023

Natural Language Processing (AI5402)

Time-1:30 Hours

Max. Marks-20

Note: Answer all questions. Write pseudo codes for all algorithms asked.
Assume suitable missing data, if any.

Q1. Consider the following Document Id, along with keywords and final class label where first six documents are training sample and last document is a test sample.

	Doc Id	Keyword	Label
Training	1	Love, Happy, Joy, Joy, Happy	Yes
	2	Happy, Love, Kick, Joy, Happy	Yes
	3	Love, Move, Joy, Good	Yes
	4	Joy, Love, Pain, Kick, Pain	No
	5	Pain, Pain, Love, Kick	No
	6	Love, Happy, Joy, Love, Pain	Yes
Test	7	Love, Pain, Joy, Love, Kick	?

Using the notion of Naïve Bayes Classification method, identify test sample document (Doc Id 7) will be classified as Yes, or No? Write all mathematical steps. [5M]

Q2. (a) Distinguish between Stemming and Lemmatization in NLP pre-processing steps? [2M]

(b) Compute the TF-IDF for the various words in the following three sentences.

S1: It is going to rain today

S2: Today I am not going outside

S3: I am going to watch the season premiere [3M]

Q3. Compute the minimum edit distance using dynamic programming approach between two strings S1= SATURDAY, and S2= SUNDAYS. Assume the cost of an insertion and deletion operation as 1 unit each and cost of a substitution operation as 2 units. Write the suitable recurrence relation for the same and show all calculation steps. [5M]

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Q4. (a) Write in general four challenges for processing low resource languages like Hindi for various NLP applications. [2M]

(b) Consider input alphabet $\Sigma = \{0,1\}$, write regular expression for all the strings over 0 and 1 which denotes

(i) even number of 1's

(ii) two consecutive 0's and two consecutive 1's [3M]

0110
1010

0011

$(01)^*(01)^*$

$0^*(11)^*0^*$

$(1)^*0^*1^*$

$(11)^*0^*$

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Roll No. 2022/APP/20

**Second Semester
M. TECH (AI)
END TERM EXAMINATION
MAY-2023**

AI5402 Natural Language Processing

Time-3:00 Hours

Max. Marks-40

Note: Answer all questions. Write pseudo codes for all algorithms asked. Assume suitable missing data, if any. Note [CO#] refers to a particular course outcome.

Answer all questions

Q1. (a) Describe the steps of Hidden Markov Model for POS Tagging with a suitable example. [7M] [CO3]

(b) Differentiate between Rule based POS Tagging and Statistical POS Tagging. [3M] [CO3]

Q2. (a) Consider the following Document Id, along with keywords and final class label where first six documents are training sample and last document is a test sample.

	Doc Id	Keyword	Label
Training	1	Worship, Cheerful, Delight, Delight, Cheerful	Pass
	2	Cheerful, Worship, Kick, Delightful, Cheerful	Pass
	3	Worship, Move, Cheerful, Good	Pass
	4	Cheerful, Worship, Pain, Kick, Pain	Fail
	5	Pain, Pain, Worship, Kick	Fail
	6	Worship, Happy, Cheerful, Worship, Pain	Pass
Test	7	Worship, Pain, Cheerful, Worship, Kick	?

Using the notion of Naïve Bayes Classification method, identify test sample document (Doc Id 7) will be classified as Pass, or Fail? Write all mathematical steps. [7M] [CO3]

(b) Differentiate between Support Vector and K-NN classifier in the context of text classification? [3M] [CO3]

Q2. (a) Distinguish between Text summarization and Topic Modelling? [3M] [CO5]

(b) What do you mean by Latent Dirichlet Allocation (LDA) for performing topic modelling. Write all steps involved in LDA. [7M] [CO5]

Q3. (a) Compute the minimum edit distance using dynamic programming approach between two strings $S1=INTENSION$, and $S2=EXECUTION$. Assume the cost of an insertion and deletion operation as 1 unit each and cost of a substitution operation as 2 units. Write the suitable recurrence relation for the same and show all calculation steps. [8M] [CO2]

(b) Write in general four challenges for performing sentiment analysis in low resource language like Hindi. [2M] [CO5]

Q4. (a) Distinguish between extractive and abstractive text summarization [2M] [CO5]

(b) Define NER (Named Entity Recognition) and design one statistical method of your choice for performing NER with suitable example? [6M] [CO4]

(c) Consider input alphabet $\Sigma=\{a,b\}$ write regular expression for all the strings over 'a' and 'b' which denotes

(i) odd number of 'a's

(ii) three consecutive a's and two consecutive b's [2M] [CO1]